

Multiple Choice (10 marks)

1. A refracting telescope consists of two converging lenses separated by 100 cm. The eye-piece lens has a focal length of 20 cm. The angular magnification of the telescope is
 - (a) 4
 - (b) 5
 - (c) 6
 - (d) 20
2. If the absolute temperature of a blackbody is increased by a factor of 3, the energy radiated per second per unit area does which of the following?
 - (a) Decreases by a factor of 81.
 - (b) Decreases by a factor of 9.
 - (c) Increases by a factor of 9.
 - (d) Increases by a factor of 81.
3. According to the BCS theory, the attraction between Cooper pairs in a superconductor is due to
 - (a) the weak nuclear force
 - (b) the strong nuclear force
 - (c) vacuum polarization
 - (d) interactions with the ionic lattice
4. The rest mass of a particle with total energy 5.0 GeV and momentum 4.9 GeV/c is approximately
 - (a) $0.1 \text{ GeV}/c^2$
 - (b) $0.2 \text{ GeV}/c^2$
 - (c) $0.5 \text{ GeV}/c^2$
 - (d) $1.0 \text{ GeV}/c^2$
5. The negative muon, μ^- , has properties most similar to which of the following?
 6. Electron
 7. Meson
 8. Photon
 9. Boson

True / False (10 marks)

Answer True or False

6. True or False: An atom with filled $n = 1$ and $n = 2$ levels contains a total of 6 electrons.
7. True or False: For an electron in the $l = 2$ state, there are 5 allowed values of the magnetic quantum number m_l : -2, -1, 0, +1, +2.
8. True or False: A particle moving at $v=0.60 c$ will travel 450 m in the lab frame before decaying.
9. True or False: The angular magnification of a refracting telescope with a 20 cm focal length eye-piece lens is 4.
10. True or False: The resolving power of the grating spectrometer for wavelengths of 500 nm and 502 nm is 250.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the process by which characteristic X-rays are generated when high-energy electrons interact with a metal target, focusing on the role of inner shell electron vacancies.
12. Discuss the differences between bosons and fermions, focusing on the implications of their wave function symmetries and the Pauli exclusion principle on their physical behavior.

End of Paper